

Questions Related to Delegates

1. What is a delegate in C#? Explain its purpose and write the general syntax for declaring and using a delegate.
2. Create a delegate that accepts two integer values and returns their sum. Create an appropriate method and invoke it using the delegate.
3. Explain the difference between unicast and multicast delegates. Write a program demonstrating both.
4. Create a delegate for a calculator application. Use the same delegate to invoke methods for addition, subtraction, multiplication, and division.
5. A notification system can send notifications using Email, SMS, and WhatsApp. Create a multicast delegate that invokes methods representing these three notification services.
6. A company calculates an employee's salary using separate methods for the following components:
 1. Basic Salary,
 2. HRA (20 % of Basic Salary),
 3. DA (10 % of Basic Salary)
 4. Bonus (15 % of Basic Salary)Use a multicast delegate to invoke these methods. Use `GetInvocationList()` to execute each method separately and display the return value of every salary component.
7. Create a multicast delegate containing at least three methods. Use `GetInvocationList()` to display the names of all methods stored in the delegate and invoke them individually.
8. What is the built-in Action delegate? Create a program demonstrating both unicast and multicast Action delegates without using lambda expressions.
9. What is the built-in Func delegate? Write a program using Func with methods that accept input values and return a result. Also demonstrate a multicast Func delegate and use `GetInvocationList()` to invoke each method separately and display all returned values.
10. What is the built-in Predicate delegate? Create a program that uses `Predicate<int>` with different methods to check whether a number is even, positive, or greater than 100. Demonstrate multicast invocation using `GetInvocationList()` and display the result returned by each method.